2021-10585 [illegible]

6. Decide whether $x^2 \equiv 150 \pmod{1009}$ is solvable or not.

claim solvable

- 1009 is prime and $150 = 2 \times 3 \times 5^2$
- $x^2 \equiv 150 \pmod{1009}$ is solvable $\iff \left(\frac{150}{1009}\right) = 1$
- $\left(\frac{3}{1009}\right) = \left(\frac{1009}{3}\right) = \left(\frac{1}{3}\right) = 1$ $\because$ quad reciprocity $\left(\frac{3}{1009}\right)\left(\frac{1009}{3}\right) = (-1)^{\frac{1}{4}\cdot 1008 \cdot 2}$
- $\left(\frac{150}{1009}\right) = \underbrace{\left(\frac{5}{1009}\right)^2}_{=1} \cdot \left(\frac{3}{1009}\right) \cdot \underbrace{\left(\frac{2}{1009}\right)}_{=1} = 1$ $\because$ Gauss $\left(\frac{2}{1009}\right) = (-1)^{\frac{1}{8}(1009^2-1)}$

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7. Find all primes p such that $x^2 \equiv 13 \pmod{p}$ has a solution.

first note that $p = 2$ then $1^2 \equiv 13 \pmod{13}$

$x^2 = 13 \pmod{p}$ has solution $\overset{\text{def}}{\iff} \left(\frac{13}{p}\right) = 1$ with $\left(\frac{13}{p}\right)\left(\frac{p}{13}\right) = (-1)^{\frac{1}{4}\cdot 12 \cdot (p-1)}$

$= (-1)^{3\underbrace{(p-1)}_{\text{짝수}}}$

$= 1$

$\overset{p \text{ is odd}}{\iff} \left(\frac{p}{13}\right) = 1$

2 is primitive root of 13 $\because$ with $2^2 \cdot 3 = 12 = 13 - 1$

quad residue be congruent to one of these :

$1 \not\equiv 2^6 \equiv -1 \pmod{13}$

$1 \not\equiv 2^9 \equiv 3 \pmod{13}$

$2^{12} \equiv 1 \pmod{13}$

2^0	2^2	2^4	2^6	2^8	2^{10}
$\parallel$	$\parallel$	$\parallel$	$\parallel$	$\parallel$	$\parallel$
1	4	3	-1	-4	-3

i.e $\left(\frac{p}{13}\right) = 1$ iff $p \equiv \pm 1, \pm 3, \pm 4 \pmod{13}$

by Dirichlet $(\pm 1, 13)$ infinitely many primes in each 6 residual class

since $= (\pm 3, 13)$

$= (\pm 4, 13) = 1$

***13.** Prove that there are infinitely many primes of each of the forms $3n + 1$ and $3n - 1$. (H)

Claim $3n-1$

- otherwise residual class $(-1) \pmod 3$ only has prime $p_1 \cdots p_k$

- consider $N = 3(p_1 \cdots p_k) - 1$

(i) products of primes inside residue class 0 or $1 \pmod 3 \equiv 0$ or 1

$\Rightarrow p_j \mid N$ with $1 \le j \le k$.

(ii) however $N \equiv -1 \pmod{p_j}$ $\rightarrow\leftarrow$

Claim $3n+1$

- otherwise residual class $1 \pmod 3$ contains only $p_1 \cdots p_k$

- consider $N = (2p_1 \cdots p_k)^2 + 3$ (N is odd)

with odd prime $q \mid N$ $\quad (2p_1 \cdots p_k)^2 \equiv -3 \pmod q$

$$\Rightarrow \left(\frac{-3}{q}\right) = 1 = \begin{cases} 1 & \text{if } q \equiv 1 \pmod 6 \\ \cancel{-1 \quad \text{if } q \equiv -1 \pmod 6} \end{cases}$$

$\Rightarrow q \not\equiv -1 \pmod 6$ i.e $p_j \mid N$

$q \equiv 1 \pmod 6$

however $N \equiv 3 \pmod{p_j}$

$\not\equiv 0$ because $p_j \neq 3$

14. Let p and q be *twin primes*, that is, primes satisfying $q = p + 2$. Prove that there is an integer a such that $p|(a^2 - q)$ if and only if there is an integer b such that $q|(b^2 - p)$. (There is a famous unsolved problem to prove that the number of pairs of twin primes is infinite. What *is* known is that the sum of the reciprocals of all twin primes is, if not a finite sum, certainly a convergent series; this result can be contrasted with Theorem 1.19. A proof of this result can be found in Chapter 15 of the book by Hans Rademacher, or in Chapter 6 of the 1977 book by W. J. LeVeque listed in the General References.)

Note - 2 is not included in twin primes pair

$\exists a \in \mathbb{Z}$ $\qquad$ $\exists a \in \mathbb{Z}$

$p \mid a^2 - q$ iff $a^2 \equiv 2 \pmod{p}$ iff $\left(\frac{2}{p}\right) = 1$

$\exists b \in \mathbb{Z}$ $\qquad$ $\exists b \in \mathbb{Z}$

$q \mid b^2 - p$ iff $b^2 \equiv -2 \pmod{q}$ iff $\left(\frac{-2}{q}\right) = 1$

$$\left(\frac{2}{p}\right)\left(\frac{-2}{q}\right) = \left(\frac{2}{p}\right)\left(\frac{2}{q}\right)\left(\frac{-1}{q}\right)$$

$$= (-1)^{\frac{1}{8}(p^2-1)} (-1)^{\frac{1}{8}(q^2-1)} (-1)^{\frac{1}{2}(q-1)}$$

$$= (-1)^{\frac{1}{8}(p^2+q^2-6+4q)} = (-1)^{\frac{1}{4}\overbrace{(q+1)}^{\text{even}}\overbrace{(q-1)}^{\text{even}}} = 1$$

either $4 \mid q+1$

$4 \mid q-1$

$$q^2 + (q-2)^2 - 6 + 4q = 2q^2 - 2$$

$$= 2(q+1)(q-1)$$

***16.** Show that if $p = 2^{2^n} + 1$ is prime then 3 is a primitive root (mod p) and that 5 and 7 are primitive roots provided that $n > 1$.

$\nu = \frac{1}{2}(p-1)$

n is primitive root (mod p) $\iff$ $(p-1)$의 모든 소인수 q에 대하여 (only $2 \mid 2^{2^n}$)

$$n^{(p-1)/q} \not\equiv 1 \pmod p$$

$$\iff \underbrace{-1}_{\pm 1 \text{ by Fermat}} \equiv (n)^{\nu} \equiv \left(\frac{n}{p}\right) \pmod p$$

$$\left(\frac{3}{p}\right) = \left(\frac{3}{2^{2^n}+1}\right) = \left(\frac{2^{2^n}+1}{3}\right) = \left(\frac{-1}{3}\right) = -1$$

$(n \geq 1)$ $(4)^{2^{n-1}} \equiv 0 \pmod 4$

$(n \geq 1)$ $(4)^{2^{n-1}} \equiv 1 \pmod 3$

$$\left(\frac{5}{p}\right) = \left(\frac{5}{2^{2^n}+1}\right) = \left(\frac{2^{2^n}+1}{5}\right) = \left(\frac{2}{5}\right) = (-1)^{\frac{1}{8}\cdot(25-1)} = (-1)^3 = -1$$

Gauss Lemma.

$(n \geq 2)$ $(16)^{2^{n-2}} \equiv 0 \pmod 4$

$(n \geq 2)$ $(16)^{2^{n-2}} \equiv 1 \pmod 5$

$$\left(\frac{7}{p}\right) = \left(\frac{7}{2^{2^n}+1}\right) = \left(\frac{2^{2^n}+1}{7}\right) =$$

$$\left(\frac{3}{7}\right) = -\left(\frac{7}{3}\right) = -\left(\frac{1}{3}\right) = -1$$

$$\left(\frac{5}{7}\right) = \left(\frac{7}{5}\right) = \left(\frac{2}{5}\right) = (-1)^{\frac{1}{8}\cdot 24} = -1$$

$(n \geq 2)$ $(16)^{2^{n-2}} \equiv 0 \pmod 4$

$2^n \pmod 3$	~~0~~	1	2
$2^{2^n} \pmod 7$	~~1~~	2	4.

$\frac{k}{2}$가능

4. Determine whether $x^4 \equiv 25 \pmod{1013}$ is solvable, given that 1013 is a prime.

Not Solvable!

i.e suffice to check

$x^4 \equiv 25 \pmod{1013}$ iff $x^2 \equiv \underbrace{\pm 5}_{\text{by Lagrange.}} \pmod{1013}$

$\left(\frac{5}{1013}\right) = 1$ or $\left(\frac{-5}{1013}\right) = 1$

$$\left(\frac{5}{1013}\right) = \left(\frac{1013}{5}\right) = \left(\frac{3}{5}\right) = \left(\frac{5}{3}\right) = \left(\frac{-1}{3}\right) = -1$$

$$\left(\frac{-1}{1013}\right) = \underbrace{(-1)^{\frac{1}{2}(1012)}}_{\text{by Euler}} = 1 \quad \text{with} \quad \left(\frac{-5}{1013}\right) = -1$$

***14.** Suppose that p is a prime, $p \equiv 1 \pmod 4$, and that $a^2 + b^2 = p$ with a odd and positive. Show that $\left(\frac{a}{p}\right) = 1$.

$\left(\frac{a}{p}\right) = \left(\frac{p}{a}\right)$ because $\frac{1}{4}(p-1)(a-1) = \frac{1}{4} \cdot 4k \cdot (a-1)$

$= k \cdot \underbrace{(a-1)}_{\text{짝수}} \equiv 0 \pmod 2$

why?

$$\left(\frac{p}{a}\right) = \left(\frac{a^2+b^2}{a}\right) = \left(\frac{b^2}{a}\right) = \left(\frac{b}{a}\right)^2 = 1$$

$\because$ $(a, b) = 1$, otherwise

$1 \neq (a, b) \mid a^2 + b^2 = p$

9. For which positive integers n do there exist integers x and y with $(x, n) = 1$, $(y, n) = 1$, such that $x^2 + y^2 \equiv 0 \pmod{n}$?

$x^2+y^2 \equiv 0 \pmod{n}$ iff $(xy^{-1})^2 + 1 \equiv 0 \pmod{n}$

iff $(xy^{-1})^2 \equiv -1 \pmod{n}$

iff $z^2 \equiv -1 \pmod{n}$ ~~with $(z, n) = 1$~~

implied by $z^2 \equiv -1$

$\Longleftrightarrow$ (CRT) $n = 2^a \cdot p_1^{e_1} \cdots p_k^{e_k} q_1^{f_1} \cdots q_m^{f_m}$

$p \equiv 1 \ (4)$ $\quad$ $q \equiv 3 \ (4)$

then $z^2 \equiv -1 \pmod{2^a}$ $\Longleftrightarrow$ $n = 2 \cdot q_1^{f_1} \cdots q_m^{f_m}$

$z^2 \equiv -1 \pmod{p_i^{e_i}}$

$z^2 \equiv -1 \pmod{q_j^{e_j}}$

where prime $q_j \equiv 3 \pmod{4}$

and $0 \le f_i \in \mathbb{Z}$

$z^2 \equiv -1 \pmod{q}$ 는 해가 없다. $\because$ otherwise $r = \frac{1}{2}(q-1)$ (홀수)

$z^{2r} = (-1)^r = -1$ $\quad$ Fermat $\rightarrow\leftarrow$

$z^2 \equiv -1 \pmod{2} \Rightarrow z \equiv 1 \pmod{2}$

$z^2 \equiv -1 \pmod{4} \Rightarrow$ 해가 없다 $(z^2 \equiv 0$ or $1)$

$z^2 \equiv -1 \pmod{p} \Rightarrow$ with $r = \frac{1}{2}(q-1)$ (홀수)

$(1 \cdots r)((-r) \cdots (-1)) \equiv -1 \pmod{p}$

$z = \pm(r!)$ when $r = \frac{1}{2}(p-1)$

$z^2 \equiv -1 \pmod{p^f} \Rightarrow$ solvable by Hensel $\because$ $(z^2+1)' = 2z \not\equiv 0$

coprime with p

Theorem 3.7 *If Q is odd and $Q > 0$, then*

$$\left(\frac{-1}{Q}\right) = (-1)^{(Q-1)/2} \quad \text{and} \quad \left(\frac{2}{Q}\right) = (-1)^{(Q^2-1)/8}$$

let $Q = p_1 \cdots p_k$

(홀수 곱수, 중복 허용)

$$\left(\frac{-1}{Q}\right) = \left(\frac{-1}{p_1}\right) \cdots \left(\frac{-1}{p_k}\right)$$

$$= (-1)^{\frac{1}{2}(p_1-1)} \cdots (-1)^{\frac{1}{2}(p_k-1)}$$

by induction $\sum \frac{1}{2}(p_j-1) \equiv \frac{1}{2}(Q-1)$ (mod 2)

Note. 홀수 a, b

$$(a-1)(b-1) \equiv 0 \pmod 4$$

$$(\underbrace{ab}_{\text{홀수}}-1) \equiv (a-1)+(b-1) \pmod 4$$

$$\frac{1}{2}(ab-1) \equiv \frac{1}{2}(a-1) + \frac{1}{2}(b-1) \pmod 2$$

$$\left(\frac{2}{Q}\right) = \left(\frac{2}{p_1}\right) \cdots \left(\frac{2}{p_k}\right)$$

$$= (-1)^{\frac{1}{8}(p_1^2-1)} \cdots (-1)^{\frac{1}{8}(p_k^2-1)}$$

$$= (-1)^{\frac{1}{8}(Q^2-1)}$$

Note. 홀수 a, b

$$(a^2-1)(b^2-1) \equiv 0 \pmod{16}$$

$$(a^2b^2-1) \equiv (a^2-1)(b^2-1) \pmod{16}$$

$$\frac{1}{8}(\underbrace{a^2b^2}_{\text{홀수}}-1) \equiv \frac{1}{8}(a^2-1)(b^2-1) \pmod 2$$

Theorem 3.8 *If P and Q are odd and positive and if $(P, Q) = 1$, then*

$$\left(\frac{P}{Q}\right)\left(\frac{Q}{P}\right) = (-1)^{\{(P-1)/2\}\{(Q-1)/2\}}.$$

$P = p_1 \cdots p_k$ (홀수 허용)

$Q = q_1 \cdots q_l$

then $\left(\frac{P}{Q}\right)\left(\frac{Q}{P}\right) = \prod_{(i,j)} \left(\frac{p_i}{q_j}\right)\left(\frac{q_j}{p_i}\right) = \prod_{(i,j)} (-1)^{\frac{1}{4}(p_i-1)(q_i-1)}$

$$\frac{1}{2}\sum_{i=1}^{k}(p_i-1) \equiv \frac{1}{2}(P-1) \pmod 2$$

$$\frac{1}{2}\sum_{j=1}^{l}(q_j-1) \equiv \frac{1}{2}(Q-1) \pmod 2$$

$$\Rightarrow \frac{1}{4}(P-1)(Q-1) \equiv \frac{1}{4}\left(\sum_{i=1}^{k} p_i - 1\right)\left(\sum_{j=1}^{l} q_j - 1\right)$$

$$\equiv \frac{1}{4}\left(\sum_{i=1}^{k}\sum_{j=1}^{l}(p_i-1)(q_j-1)\right)$$

2. Prove that the quadratic form $x^2 - 2xy + y^2$ has discriminant 0. Determine the class of integers represented by this form.

$$F = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \qquad \det F = 1^2 - (-1)(-1) = 0 = ac - \frac{b^2}{2}$$

$f = X^T F X$

where $X = \begin{pmatrix} x \\ y \end{pmatrix}$ hence discriminant $= 0$

- $f = (x-y)^2$ i.e f represents set of perfect squares

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4. Use the binomial theorem to give a formula for positive integers x_k and y_k such that $(3+2\sqrt{2})^k = x_k + y_k\sqrt{2}$. Show that $(3-2\sqrt{2})^k = x_k - y_k\sqrt{2}$. Deduce that $x_k^2 - 2y_k^2 = 1$ for $k = 1, 2, 3, \cdots$. Show that $(x_k, y_k) = 1$ for each k. Show that $x_{k+1} = 3x_k + 4y_k$ and $y_{k+1} = 2x_k + 3y_k$ for $k = 1, 2, 3 \cdots$. Show that $\{x_k\}$ and $\{y_k\}$ are strictly increasing sequences. Conclude that the number 1 has infinitely many proper representations by the quadratic form $x^2 - 2y^2$.

x_k $\qquad$ $y_k\sqrt{2}$

① $$(3+2\sqrt{2})^k = \boxed{\sum_{\substack{j=0 \\ j \text{ 는 짝수}}}^{k} \binom{k}{j} \cdot \underbrace{(2\sqrt{2})^j}_{(8)^{j/2}} \, 3^{k-j}} + \boxed{\sum_{\substack{j=0 \\ j \text{ 는 홀수}}}^{k} \binom{k}{j} \cdot \underbrace{(2\sqrt{2})^j}_{(2\sqrt{2})(8)^{(j-1)/2}} \, 3^{k-j}}$$

$$(3-2\sqrt{2})^k = \boxed{\sum_{\substack{j=0 \\ j \text{ 는 짝수}}}^{k} \binom{k}{j} \cdot \underbrace{(2\sqrt{2})^j}_{(8)^{j/2}} \, 3^{k-j}} - \boxed{\sum_{\substack{j=0 \\ j \text{ 는 홀수}}}^{k} \binom{k}{j} \cdot \underbrace{(2\sqrt{2})^j}_{(2\sqrt{2})(8)^{(j-1)/2}} \, 3^{k-j}}$$

② $(3+2\sqrt{2})^k (3-2\sqrt{2})^k = (9-8)^k = x_k^2 - 2y_k^2 = 1$

(x_k, y_k) divides $1 \implies (x_k, y_k) = 1$

③ $x_{k+1} + \sqrt{2}\, y_{k+1} = (3+2\sqrt{2})(x_k + \sqrt{2}\, y_k)$

$= (3x_k + 4y_k) + (2x_k + 3y_k)\sqrt{2}$

strict increasing!

④ $x_1 = 3$, $y_1 = 2$ therefore by induction $x_k, y_k > 0$ with $\Delta x_k = 2x_k + 4y_k > 0$, $\Delta y_k = 2x_k + 2y_k > 0$

1. Find a reduced form that is equivalent to the form $7x^2 + 25xy + 23y^2$.

$(a, b, c) \to (c, -b, a)$

$\to (a, b+2a, a+b+c)$

$\to (a, b-2a, a-b+c)$

$(7, 25, 23) \xrightarrow{③} (7, 11, 5) \xrightarrow{①} (7, -3, 1)$

$\xrightarrow{⑥} (1, 3, 7)$ $\quad -a < b \le a < c$

$\xrightarrow{①} (1, 1, 5)$ $\quad x^2 + xy + 5y^2$

---

3. Let x and y be integers. Show that there exist integers u and v such that $\begin{bmatrix} x & y \\ u & v \end{bmatrix} \in \Gamma$ if and only if $(x, y) = 1$.

by Bézout's identity

$\Rightarrow$ then $\det \begin{bmatrix} x & y \\ u & v \end{bmatrix} = xv - uy = 1$ $\quad (x, y) = 1$

$\Leftarrow$ $(x, y) = 1$ then $xv - uy = 1$ $\exists\, v, u \in \mathbb{Z}$ then $\begin{bmatrix} x & y \\ u & v \end{bmatrix} \in \Gamma$

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4. Show that a binary quadratic form f properly represents an integer n if and only if there is a form equivalent to f in which the coefficient of x^2 is n. Use this and (3.3) to give a second proof of Theorem 3.13.

U represents integral unimodular substitution

$(\Rightarrow)$ $f(p, r) = n$ then $\exists\, q, s \in \mathbb{Z}$ s.t. $pq - rs = 1$

with $(p, r) = 1$

$F' = U^T F U$

then coefficient of x^2 in $f' = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T F' \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ r \end{pmatrix}^T F \begin{pmatrix} p \\ r \end{pmatrix} = 1$

(f' equivalent to f)

$(\Leftarrow)$ if coefficient of $f' = n$ then $f'(1, 0) = n$

then $\begin{pmatrix} 1 \\ 0 \end{pmatrix}^T F' \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}^T U^T F U \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ r \end{pmatrix}^T F \begin{pmatrix} p \\ r \end{pmatrix}$

$= f(p, r)$ with $pq - rs = 1$

5. Show that $x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$ are the only reduced quadratic forms of discriminant -20. Show that the first of these forms does not represent 2, but that the second one does. Deduce that these forms are inequivalent, and hence that $H(-20) = 2$. Show that an odd prime p is represented by at least one of these forms if and only if $p \equiv 1, 3, 7$, or $9 \pmod{20}$.

- $-20 = b^2 - 4ac \le a^2 - 4a^2$ i.e $a = 1$ or 2

  $= -3a^2$

- b는 짝수. $-20 \equiv b^2 - 4ac \pmod 2$

- case ① $a=1$

  $b=0$. $\Rightarrow$ $-20 = -4c$

  $c=5$

- case ② $a=2$

  $b = \cancel{-2}, 0, 2$

  기약

  조건 위배

  $b=0$ then $-20 = -8c$ $\nexists c$

  $b=2$ then $-20 = 4 - 8c$ $c=3$

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x^2+5y^2 if $y \neq 0$ then $x^2+5y^2 \ge 5 > 2$

$x=0$

$y=1$

if $y=0$ then $x^2 = 2$ (불가능)

$2x^2 + 2xy + 3y^2$. with $x=1$, $y=0$ $2 \cdot 1^2 + 2 \cdot 1 \cdot 0 + 3 \cdot 0^2 = 2$.

$(1, 0) = 1$

$\Rightarrow$ 2를 표현할 수 있으나 아니다

같지 않기 때문에 두 식은 inequivalent

odd p

is representable iff $\left(\frac{-20}{p}\right) = 1$

by quad form with d

$$\left(\frac{-20}{p}\right) = \left(\frac{-1}{p}\right)\underbrace{\left(\frac{4}{p}\right)}_{=1}\left(\frac{5}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{p}{5}\right)$$

$$\begin{cases} +1 & p \equiv 1 \text{ or } 4 \pmod 5 \\ -1 & p \equiv 2 \text{ or } 3 \pmod 5 \end{cases}$$

$$\begin{cases} +1 & p \equiv 1 \pmod 4 \\ -1 & p \equiv 3 \pmod 4 \end{cases}$$

$p \equiv 1, 9 \pmod{20}$

$p \equiv 3, 7 \pmod{20}$

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6. Let $f(x, y) = ax^2 + bxy + cy^2$ and $g(x, y) = f(-x, y) = ax^2 - bxy + cy^2$. These forms represent precisely the same numbers, but they are not necessarily equivalent (because the determinant of the transformation has determinant -1). Show that $x^2 + xy + 2y^2$ is equivalent to $x^2 - xy + 2y^2$, but that $3x^2 + xy + 4y^2$ and $3x^2 - xy + 4y^2$ are not equivalent.

$x^2 + xy + 2y^2$ is not properly reduced!

$(-a = b)$

$$F \qquad U$$

$$\begin{bmatrix} 1 & 1/2 \\ 1/2 & 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 \\ & 1 \end{bmatrix}$$

$\det U = 1$

then

$$G = U^T F U = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1/2 \\ 1/2 & 3/2 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 2 \end{bmatrix}$$

$3x^2 + xy + 4y^2 \Rightarrow$ satisfies $-a < b \le a < c$

$-3 < 1 \le 3 < 4$

$3x^2 - xy + 4y^2 \Rightarrow$ satisfies $-a < b \le a < c$

$-3 < -1 \le 3 < 4$

both are already reduced forms $\Rightarrow$ not equivalent

7. Let $f(x, y)$ be a quadratic form whose discriminant d is a positive perfect square. Show that f is equivalent to a form $ax^2 + bxy + cy^2$ for which $c = 0$ and $0 \leqslant a < |b|$. Deduce that there are only finitely many equivalence classes of forms of this discriminant. (H)

$$4af(x,y) = (2ax+by)^2 - dy^2$$

$$= (2ax + (b+\sqrt{d})y)\ (2ax + (b-\sqrt{d})y)$$

$$\frac{\xi}{s} = \frac{-b \pm \sqrt{d}}{2a} \quad \text{then} \quad f(\xi, s) = 0$$

with $(\xi, s) = 1 \rightsquigarrow \exists\, p, r \in \mathbb{Z}$

s.t. $p\xi - rs = 1$

$U_1 = \begin{pmatrix} p & \xi \\ r & s \end{pmatrix}$ then $U_1^T F U_1 = F_1$ $\qquad F_1 = \begin{bmatrix} A & B/2 \\ B/2 & 0 \end{bmatrix}$

$\det U = 1$

$f_1(0,1) = 0$

$f_1 = Ax^2 + Bxy$

$U_2 = \begin{pmatrix} 1 & 0 \\ m & 1 \end{pmatrix}$ then $U_2^T F_1 U_2 = F_2$ $\qquad F_2 = \begin{bmatrix} A+Bm & B/2 \\ B/2 & 0 \end{bmatrix}$

$\det U' = 1$

$f_2 = (A+Bm)x^2 + Bxy$

(any $m \in \mathbb{Z}$)

let $A = -mB + r \quad (0 \leq r < |B|)$

then $f_2 = rx^2 + Bxy$

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of b : $b^2 - 4a \cdot 0 = d$, $b = \pm\sqrt{d}$ (2 ways)

of a : $0 \leq a < |b| = \sqrt{d}$

hence less than $2 \cdot \sqrt{d}$ ways